

Vector Calculus Exercises

Chapter 2.1

Exercise 11

Given the set of points $\{(x_i, y_i)\}_{i=1}^n$, and the function $f(m, b)$ defined as:

$$f(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2,$$

Minimize $f(m, b)$ using the method of least squares to find the best fit line $y = mx + b$ for the given data points.

We first compute the partial derivatives of $f(m, b)$ with respect to m and b , and set them to zero to find the critical points.:

$$\frac{\partial f}{\partial m} = -2 \sum_{i=1}^n x_i (y_i - mx_i - b) = 0,$$

$$\frac{\partial f}{\partial b} = -2 \sum_{i=1}^n (y_i - mx_i - b) = 0.$$

Solving the second equation for b gives:

$$b = \frac{1}{n} \sum_{i=1}^n y_i - m \frac{1}{n} \sum_{i=1}^n x_i = \bar{y} - m\bar{x},$$

where $\bar{x}$ and $\bar{y}$ are the means of the x_i and y_i values, respectively. Substituting this expression for b back into the first equation, we can solve for m :

$$-2 \sum_{i=1}^n x_i (y_i - mx_i - (\bar{y} - m\bar{x})) = 0.$$

Expanding and simplifying this equation leads to:

$$\sum_{i=1}^n x_i y_i - m \sum_{i=1}^n x_i^2 - \bar{y} \sum_{i=1}^n x_i + m\bar{x} \sum_{i=1}^n x_i = 0.$$

Then we can rearrange this to find m :

$$m = \frac{\sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2 - \bar{x} \sum_{i=1}^n x_i}.$$

Otherwise, we can express y_i in terms of the best fit line:

$$y_i = mx_i + b,$$

and form the matrix equation:

$$A \begin{bmatrix} m \\ b \end{bmatrix} = \mathbf{y},$$

where A is the matrix of x_i values and a column of ones, and $\mathbf{y}$ is the vector of y_i values.

This is an inconsistent system, although there exists a solution:

$$\begin{bmatrix} \hat{m} \\ \hat{b} \end{bmatrix},$$

such that:

$$\|Y - A\hat{X}\|^2 \leq \|Y - AX\|^2 \quad \text{for all } X,$$

where $\hat{Y} \in \text{Col}(A)$ and $Y - \hat{Y} \in (\text{Col}(A))^\perp$.

Then, $A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$ such that

$$a_i^T(Y - A\hat{X}) = 0 \quad \text{for } i = 1, 2.$$

The expression above can be rewritten as:

$$A^T(Y - A\hat{X}) = 0 \implies A^TY = A^TA\hat{X} \implies \hat{X} = (A^TA)^{-1}A^TY,$$

i.e., the normal equations for least squares. Writing

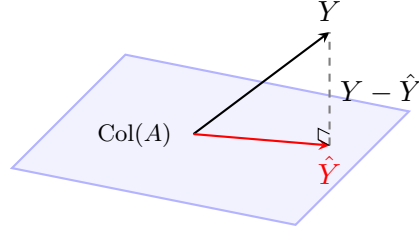
$$F(X) = f(m, b) = \|Y - AX\|^2 = (Y - AX)^T(Y - AX) = Y^TY - Y^TAX - X^TA^TY + X^TA^TAX,$$

we can compute the gradient of F as:

$$\nabla F(X) = -2A^TY + 2A^TAX.$$

Setting $\nabla F(X) = 0$ gives us the normal equations again:

$$A^TAX = A^TY \implies X = (A^TA)^{-1}A^TY.$$



Exercise 16

Given the function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ differentiable around -1 such that $h(-1) = 1$ and,

$$f(x, y) = h(xy) + 2h\left(\frac{y}{x}\right) - 4, \quad \text{for } x \neq 0,$$

find the gradient of f at the point $(-1, 1)$. Then, obtain the first-order Taylor expansion of f around the point $(-1, 1)$.

To find the gradient of f at the point $(-1, 1)$, we first compute the partial derivatives of f with respect to x and y . Using the chain rule, we have:

$$\frac{\partial f}{\partial x} = yh'(xy) - \frac{2y}{x^2} \cdot h'\left(\frac{y}{x}\right),$$

$$\frac{\partial f}{\partial y} = xh'(xy) + \left(\frac{2}{x}\right) \cdot h'\left(\frac{y}{x}\right).$$

Evaluating these derivatives at the point $(-1, 1)$, we get:

$$\frac{\partial f}{\partial x}(-1, 1) = 1 \cdot h'(-1) - \frac{2 \cdot 1}{(-1)^2} \cdot h'(-1) = h'(-1) - 2h'(-1) = -h'(-1),$$

$$\frac{\partial f}{\partial y}(-1, 1) = -1 \cdot h'(-1) + \left(\frac{2}{-1}\right) \cdot h'(-1) = -h'(-1) - 2h'(-1) = -3h'(-1).$$

Since $h'(-1) = 1$, the gradient of f at the point $(-1, 1)$ is:

$$\nabla f(-1, 1) = (-1 \quad -3).$$

The first-order Taylor expansion of f around the point $(-1, 1)$ is given by:

$$\mathcal{P}_{1,(-1,1)}(x, y) = f(-1, 1) + \nabla f(-1, 1) \cdot \begin{pmatrix} x + 1 \\ y - 1 \end{pmatrix}.$$

Substituting the values, we have:

$$\mathcal{P}_{1,(-1,1)}(x, y) = -1 - h'(-1)(x + 1) - 3h'(-1)(y - 1) = -1 - (x + 1) - 3(y - 1)$$